

PetAffect

Uncertainty-aware audio modeling of companion-animal welfare signals

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Research focus

A long-term, audio-first research program investigating whether carefully evaluated machine learning can surface structured acoustic observations and potential within-animal change for human review, without claims of translation, diagnosis, or autonomous decision-making.

Initial scope

Cat-focused audio research · Consent-based data design · Calibrated uncertainty · Human oversight

The Problem

Companion animals cannot directly report discomfort, environmental distress, fear, or meaningful behavioral change. Caregivers, veterinary teams, shelter staff, boarding providers, groomers, and animal-welfare researchers must often make judgments from incomplete observations: a changed vocalization pattern, a new growl or whine, prolonged meowing, altered purring behavior, or a departure from an individual animal's typical acoustic baseline.

Those observations can be valuable, but they are inherently context-dependent. A vocalization may reflect many possible conditions, and recording quality, proximity to the microphone, background noise, and individual variation can substantially affect interpretation. In practical settings, staff may not have time to review every recording or compare an animal's behavior across days.

PetAffect will investigate whether audio models can surface structured acoustic observations and possible changes for human review. It will not attempt to infer hidden thoughts, translate animal language into English, or make autonomous welfare decisions.

Research Hypothesis

The initial research question is:

Can an uncertainty-aware audio model estimate coarse, observable vocalization and behavioral-context indicators for cats, and identify deviations from an individual animal's historical acoustic baseline, while generalizing to animals, households, sessions, and recording conditions not seen during training?

The project will begin with cats only. This narrower scope will make it possible to focus on annotation quality, recording variation, acoustic representation learning, and the practical reliability of uncertainty-aware outputs before expanding into multimodal work.

PetAffect will investigate cautiously framed outputs such as vocalization type or acoustic characteristics, low versus high acoustic arousal indicators, possible alert or defensive vocal patterns, social or interaction context indicators, insufficient evidence, and within-animal change from baseline. These will remain research estimates of observable patterns, not determinations of emotional state, pain, disease, aggression, or intent.

System Design

The initial system will center on **PetAudio**, an audio-learning component for analyzing short recordings. It will examine vocalizations and acoustic features such as meows, purrs, growls, whines, pitch, duration, rhythm, spectral characteristics, temporal structure, and changes relative to an individual animal's historical acoustic baseline.

PetAudio will pair acoustic preprocessing and learned representations with recording-quality estimation and an uncertainty-aware output layer. The system will produce coarse acoustic-context estimates, a possible

within-animal baseline-deviation signal where sufficient history exists, and an insufficient-evidence or human-review cue when reliable estimation is not supported.

The architecture will include quality checks and missing-context awareness rather than assuming every clip is usable. Low signal-to-noise ratio, overlapping human speech, multiple animals, clipping, uncertain source attribution, or recordings unlike the training distribution may trigger low-confidence outputs or abstention.

Where a sufficient historical record exists, a longitudinal component will compare current acoustic embeddings and engineered features with an individual animal's own reference distribution. This may help identify an atypical vocalization pattern relative to that animal's history, even when the pattern is not unusual across the wider population.

Research Methodology

The initial dataset will be consent-based and privacy-conscious, composed of short cat audio sessions with optional structured context. Data records may include anonymous animal IDs, session IDs, setting labels, recording-device and quality metadata, time-related metadata, and carefully minimized contextual information.

The project will use multiple independent annotations for cautiously defined labels. Potential categories will include:

- Rest or neutral acoustic context
- Low versus high acoustic arousal indicators
- Alert, avoidant, or defensive vocalization indicators
- Social or play context indicators
- Vocalization class or acoustic event type
- Insufficient evidence or abstain
- Within-animal deviation from historical baseline

Label design will prioritize observable descriptions over presumed internal states. Annotators may label what can be heard, such as repeated vocalization, growl-like acoustic pattern, vocalization duration, quietness, or context cues available in the recording, while uncertainty and disagreement are retained as information rather than discarded.

Data splitting will protect against leakage. Training, validation, and testing will be separated by animal identity, household, recording session, environment, and, where possible, device type. This will reduce the risk that a model succeeds by memorizing a particular cat, home acoustics, caregiver voice, or microphone profile rather than learning transferable acoustic relationships.

The model-development plan will compare interpretable feature-based baselines with pretrained audio encoders and fine-tuned sequence models. Candidate approaches may include mel-spectrogram models, self-supervised audio representations, temporal pooling architectures, and models that combine learned

embeddings with quality and structured metadata. The work will evaluate whether audio representations improve over transparent handcrafted baselines without sacrificing interpretability or reliability.

Evaluation and Safety

PetAffect will prioritize evidence quality over a single accuracy number. Planned evaluation will include macro F1, per-class recall, precision-recall behavior, calibration metrics, expected calibration error, risk-coverage curves, abstention performance, cross-animal generalization, and robustness to background noise, device variation, compression, overlap, and truncated recordings.

Calibration will be a core research objective. A model that supplies a confidence score is not necessarily well calibrated: a prediction reported at high confidence should be correct at approximately that rate over comparable examples. PetAffect will investigate calibration methods, selective prediction, out-of-distribution detection, and conservative abstention policies so that uncertain clips are routed for review rather than presented with false precision.

Longitudinal anomaly detection will be evaluated separately from population-level classification. The research question will be whether an audio model can identify a meaningful shift from an individual animal's normal acoustic baseline, not whether it can identify a cause. Any output will be framed as a possible behavioral change requiring human review.

What this is not: PetAffect will not be an animal-language translator, a system that reads animals' thoughts, or a tool that converts cat vocalizations into English. It will not diagnose pain, disease, aggression, or emotional states, and it will not replace a qualified veterinarian, animal behavior professional, or caregiver's judgment. Its proposed role is limited to research estimates of observable acoustic patterns, calibrated uncertainty, and structured review cues.

Research Roadmap

Phase 1: Cat-focused audio baselines. The first phase will define a consent and data-governance process, annotation protocols, identity-separated evaluation splits, recording-quality checks, and interpretable audio baselines. It will establish a disciplined foundation for evaluating acoustic features and pretrained audio representations on narrowly scoped, observable labels.

Phase 2: Multimodal fusion and uncertainty-aware evaluation. Although the initial research scope will remain audio-only, multimodal fusion is a highly desired future direction. This phase would investigate how PetAudio could be combined with visual observations, such as posture, movement, facial action cues, ear or tail position, and environmental context, while preserving calibrated confidence, missing-modality handling, and abstention behavior. Video could offer complementary evidence when audio alone is ambiguous, but it would require its own careful collection, privacy controls, annotation framework, and identity-separated validation.

Phase 3: Longitudinal personalization and behavioral-change detection. Longitudinal modeling is also a major future research interest. This phase would investigate personalized acoustic baselines and, potentially, multimodal temporal histories to identify sustained deviations from an individual animal's normal

observable behavior. The goal would remain conservative: flagging a possible change for review, not assigning a diagnosis or asserting an underlying cause.

Open Science and Reproducibility

I am approaching PetAffect as both a machine-learning research question and a production ML systems challenge. My background in AI infrastructure, model evaluation, low-latency inference, reliable deployment, and open-source LLM inference and post-training tooling will shape the project's technical direction.

When ethical and legal constraints permit, I plan to publish dataset documentation, annotation guidance, model cards, experiment configurations, training code, benchmark definitions, and evaluation tooling. Documentation will identify intended uses, non-intended uses, known limitations, uncertainty behavior, data gaps, and deployment risks.

The project will treat data provenance, versioning, model reproducibility, observability, and failure analysis as first-class engineering concerns. Any future serving system will be designed around privacy-aware data handling, auditability, clear abstention behavior, and explicit human oversight rather than autonomous decision-making.

What Success Would Mean

Success would not mean asserting that a model understands what a cat is thinking or feeling. It would mean producing rigorous evidence about a more constrained question: whether audio-based ML can estimate carefully defined observable vocalization patterns, communicate uncertainty honestly, and identify possible individual-level acoustic changes that merit human attention.

A useful outcome would be a reproducible, carefully scoped research foundation: transparent baselines, robust identity-separated evaluation, conservative abstention policies, and documentation that makes failures as visible as successes. It could establish a credible path toward future multimodal and longitudinal research without assuming that audio alone can resolve ambiguous welfare questions.

PetAffect is a long-term research-engineering program that I plan to develop alongside graduate computer-science coursework. I welcome collaboration from animal-welfare researchers, veterinary professionals, shelters, rescues, and ML practitioners interested in responsible data collection, animal-behavior measurement, audio ML, multimodal research, and uncertainty-aware AI systems.